

.ORIG x3000

LD R6, STACK_PTR

LEA R0, PROMPT_MSG
PUTS

JSR SUB_GET_INPUT

LEA R0, NEWLINE_MSG
PUTS

JSR SUB_BUBBLE_SORT
JSR SUB_PRINT_OUTPUT

HALT

STACK_PTR .FILL x4000

PROMPT_MSG .STRINGZ "Enter 8 numbers (0-100) separated by spaces, then hit Enter:\n"

NEWLINE_MSG .STRINGZ "\nSorted Output: "

ARRAY_BASE .BLKW #8

SUB_GET_INPUT

ADD R6, R6, #-1
STR R7, R6, #0
ADD R6, R6, #-1
STR R0, R6, #0
ADD R6, R6, #-1
STR R1, R6, #0
ADD R6, R6, #-1
STR R2, R6, #0
ADD R6, R6, #-1
STR R3, R6, #0
ADD R6, R6, #-1
STR R4, R6, #0
ADD R6, R6, #-1
STR R5, R6, #0

LEA R2, ARRAY_BASE
AND R1, R1, #0
ADD R1, R1, #8

GI_NUM_LOOP

AND R3, R3, #0

GI_DIGIT_LOOP

GETC

OUT

ADD R4, R0, #-16

ADD R4, R4, #-16

BRz GI_STORE_NUM

ADD R4, R0, #-10

BRz GI_STORE_NUM

LD R4, GI_ASCII_OFFSET

ADD R0, R0, R4

BRn GI_STORE_NUM

ADD R5, R0, #-10

BRzp GI_STORE_NUM

ADD R5, R3, R3

ADD R3, R5, R5

ADD R3, R3, R3

ADD R3, R3, R5

ADD R3, R3, R0

BRnzp GI_DIGIT_LOOP

GI_STORE_NUM

STR R3, R2, #0

ADD R2, R2, #1

ADD R1, R1, #-1

BRz GI_DONE

GI_SKIP_DELIMITERS

GETC

OUT

ADD R4, R0, #-16

ADD R4, R4, #-16

BRz GI_SKIP_DELIMITERS

ADD R4, R0, #-10

BRz GI_SKIP_DELIMITERS

```
LD    R4, GI_ASCII_OFFSET
ADD   R3, R0, R4
BRnzp GI_DIGIT_LOOP
```

GI_DONE

```
LDR    R5, R6, #0
ADD    R6, R6, #1
LDR    R4, R6, #0
ADD    R6, R6, #1
LDR    R3, R6, #0
ADD    R6, R6, #1
LDR    R2, R6, #0
ADD    R6, R6, #1
LDR    R1, R6, #0
ADD    R6, R6, #1
LDR    R0, R6, #0
ADD    R6, R6, #1
LDR    R7, R6, #0
ADD    R6, R6, #1
```

RET

GI_ASCII_OFFSET .FILL #48

SUB_BUBBLE_SORT

```
ADD    R6, R6, #-1
STR    R7, R6, #0
ADD    R6, R6, #-1
STR    R0, R6, #0
ADD    R6, R6, #-1
STR    R1, R6, #0
ADD    R6, R6, #-1
STR    R2, R6, #0
ADD    R6, R6, #-1
STR    R3, R6, #0
ADD    R6, R6, #-1
STR    R4, R6, #0
ADD    R6, R6, #-1
STR    R5, R6, #0
```

```
AND    R1, R1, #0
ADD    R1, R1, #7
```

BS_OUTER_LOOP

BRz BS_DONE

ADD R2, R1, #0

LEA R3, ARRAY_BASE

BS_INNER_LOOP

LDR R4, R3, #0

LDR R5, R3, #1

NOT R0, R5

ADD R0, R0, #1

ADD R0, R4, R0

BRnz BS_NO_SWAP

STR R5, R3, #0

STR R4, R3, #1

BS_NO_SWAP

ADD R3, R3, #1

ADD R2, R2, #-1

BRp BS_INNER_LOOP

ADD R1, R1, #-1

BRnzp BS_OUTER_LOOP

BS_DONE

LDR R5, R6, #0

ADD R6, R6, #1

LDR R4, R6, #0

ADD R6, R6, #1

LDR R3, R6, #0

ADD R6, R6, #1

LDR R2, R6, #0

ADD R6, R6, #1

LDR R1, R6, #0

ADD R6, R6, #1

LDR R0, R6, #0

ADD R6, R6, #1

LDR R7, R6, #0

ADD R6, R6, #1

RET

SUB_PRINT_OUTPUT

```
ADD    R6, R6, #-1
STR     R7, R6, #0
ADD    R6, R6, #-1
STR     R0, R6, #0
ADD    R6, R6, #-1
STR     R1, R6, #0
ADD    R6, R6, #-1
STR     R2, R6, #0
ADD    R6, R6, #-1
STR     R3, R6, #0
ADD    R6, R6, #-1
STR     R4, R6, #0
ADD    R6, R6, #-1
STR     R5, R6, #0
```

```
LEA     R2, ARRAY_BASE
AND     R1, R1, #0
ADD     R1, R1, #8
```

PO_NUM_LOOP

```
LDR     R3, R2, #0
```

```
AND     R4, R4, #0
```

PO_HUNDREDS_LOOP

```
LD      R0, MINUS_100
ADD     R0, R3, R0
BRn     PO_HUNDREDS_DONE
ADD     R3, R0, #0
ADD     R4, R4, #1
BRnzp   PO_HUNDREDS_LOOP
```

PO_HUNDREDS_DONE

```
AND     R5, R5, #0
```

PO_TENS_LOOP

```
LD      R0, MINUS_10
ADD     R0, R3, R0
BRn     PO_TENS_DONE
ADD     R3, R0, #0
ADD     R5, R5, #1
BRnzp   PO_TENS_LOOP
```

PO_TENS_DONE

```
ADD     R0, R4, #0
```

BRp PO_PRINT_ALL3

ADD R0, R5, #0
BRp PO_PRINT_TO

LD R0, ASCII_BASE
ADD R0, R3, R0
OUT
BRnzp PO_PRINT_SPACE

PO_PRINT_TO
LD R0, ASCII_BASE
ADD R0, R5, R0
OUT
LD R0, ASCII_BASE
ADD R0, R3, R0
OUT
BRnzp PO_PRINT_SPACE

PO_PRINT_ALL3
LD R0, ASCII_BASE
ADD R0, R4, R0
OUT
LD R0, ASCII_BASE
ADD R0, R5, R0
OUT
LD R0, ASCII_BASE
ADD R0, R3, R0
OUT

PO_PRINT_SPACE
LD R0, ASCII_SPACE
OUT

ADD R2, R2, #1
ADD R1, R1, #-1
BRp PO_NUM_LOOP

LDR R5, R6, #0
ADD R6, R6, #1
LDR R4, R6, #0
ADD R6, R6, #1
LDR R3, R6, #0
ADD R6, R6, #1

```
LDR    R2, R6, #0
ADD    R6, R6, #1
LDR    R1, R6, #0
ADD    R6, R6, #1
LDR    R0, R6, #0
ADD    R6, R6, #1
LDR    R7, R6, #0
ADD    R6, R6, #1
```

```
RET
```

```
ASCII_BASE    .FILL    x0030
ASCII_SPACE    .FILL    x0020
MINUS_100      .FILL    #-100
MINUS_10       .FILL    #-10
```

```
.END
```

Step

Next

Finish

Run

Pause

Continue

Unhalt

☒ Follow PC

Console

Enter 8 numbers (0-100) separated by spaces, then hit Enter:
11 8 2 17 6 4 3 21

Sorted Output: 2 3 4 6 8 11 17 21
----- Halting the processor -----

Clear Input Buffer (0 characters)

Clear Output

- ☒ Newline as 0x0A
- ☐ Newline as 0x0D
- ☐ Binary (leave newlines unchanged)

